

BOOKAZOR - an Online Appointment Booking System

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Abstract— Bookazor is an appointment booking and scheduling web-based application which is used for booking appointments in the streams of parlor, hospitals and architects within a defined geographic area. This application is streamlined in an ionic basis. It is an open source SDK for hybrid mobile application development. It uses technologies like CSS, HTML, and JavaScript. Firebase plays a vital role in fetching data for appointment scheduling that helps to enhance application development effectively. It provides functionalities like analytics, database, messaging, and crash reporting which helps in focusing the users. The system includes NodeJS for storing the number of requests, each of which specifies a sequence of regions to be visited by a particular user. NodeJS server is used for offering appointments at specified times, using the table to check for availability of operatives in specified regions at specified times, and for inserting new jobs in the routes to reflect booked appointments. A scheduler periodically updates the routes e.g. by means of a simulated annealing process, to generate a new set of appointments and prevents duplication.

Keywords— Ionic, Software Development Kit(SDK), Cascading Style Sheet(CSS), HyperText Markup Language(HTML), JavaScript, Firebase, NodeJS

I. INTRODUCTION

In general every organization or line of business requires a person to book appointments, which needs to be confirmed or verified. In today's society individuals are generally on the go and appointments are made, modified and confirmed at a rapid rate, requiring patience, time and cost. Furthermore there is generally a lag between requesting an appointment time and having the appointment approved. One such area where appointments are made is in the medical field where a patient or user needs to book appointment with a family doctor who may then subsequently require the patient to book an appointment with a specialist, hospital, blood lab, X-ray clinic or other scanning facility. It is not unusual for the patient to repeat basic information all over again. There have been various attempts to improve this scheduling issue. The same scenario includes for scheduling an appointment in a parlor, event managers, household services, etc. It is an objective of this system to provide an improved online booking service and computer implemented method relating to the same. It is a further objective of the system to provide a single centralized booking service not just for patients but also for users who are in need of carpenters, plumbers, architects etc. Thus this paper provides a single platform to schedule an appointment anytime within the user's geographical region.

II. LITERATURE REVIEW

In this proposed system, the application for scheduling an appointment is defined in simple terms with respect to the design, input, process and output using NodeJS as backend and angular as frontend. The purpose of this study was to identify the benefits and barriers to implement Web-based medical scheduling discussed in the literature as well as the unmet needs under the current health care environment[1]. An intelligent agent based system towards negotiating and collaborating with the agents of doctors and the hospital for the appropriate appointment time for the patient which would take the above factors into consideration. In addition the meetings of the junior staff like the duty doctor and nurse with the chief doctor regarding patients would also be carried out again while taking into consideration the medical condition of the patient admitted and so on[2]. In this paper, the proposed system is a smart appointment booking method that provides patients an easy way of selecting a doctor's appointment online. This will promote the patient hope and satisfaction which increases the profit margin. In this system user authentication is provided. This will be an online application that overcomes the problem of handling and booking appointments according to user's choice. It is a user friendly appointment system which requires less manpower and is also secure than offline appointment. We can retrieve information whenever required[3]. For appropriate scheduling, Scheduled consultation times of physicians are entered into the Patient Appointment Reservation System by admin. There is an opportunity for specialists to enter their planned consultation times themselves if they sign into the system. Patient can view the hospital details, doctor's profile including the visiting time of the doctor. For booking an appointment the patients' details are entered into Patient Appointment Reservation System by registering themselves through web browser. After registration the patients can login and book their appointment with their preferences [4]. For enhanced appointments, a study presents a web based appointment booking system through web or mobile devices that assists both students and lecturers to be acquainted with the time of appointment wherever they are. The system allows students and lecturers to simply gain access to the system by connecting to the Internet [5]. The procedure regarding the pros and cons of setting up an appointment is simplified[6]. The dynamic scheduling without any manual schedule process is explained [7]. This paper provides an ideal design paradigm for the development of integrated health care information system in the primary care setting [8]. Considering these drawbacks, we here propose to develop an alternate patient appointment system using Near Field Communication (NFC) technique and Android enabled mobile application with a view to redefining the core of

hospital waiting time towards appointment[9]. This paper [10] proposes dentist online Reservation System which saves the time and effort of the users in safe and convenient ways. It enables patients to book and check their appointment schedule. Administrator maintains the database online easily. It enables the administrator to check the patient’s requests, manage the appointment schedule, and manage the patient’s information.

III. FRAMEWORK

The framework for this application comprises of login module for the user’s sign up, scheduler for scheduling an appointment, categories module comprises of the various domains such as events, fashion, gym, salon, health, programming, architects, a database for storing the user’s requests for any appointment and also consists of the details of the registered user.

3.1. LOGIN MODULE:

This module is the vital one which helps the users to register as customers for scheduling an appointment or register their business details into one of the categories so that other users can be directed to their business service category. As soon as the user tends to open the application, their location is determined with the permission of the user. After enabling the location, the users will be able to view the Application content but will not be able to schedule an appointment. Thus, the user needs to sign up and then login in order to schedule an appointment for the user desired category.

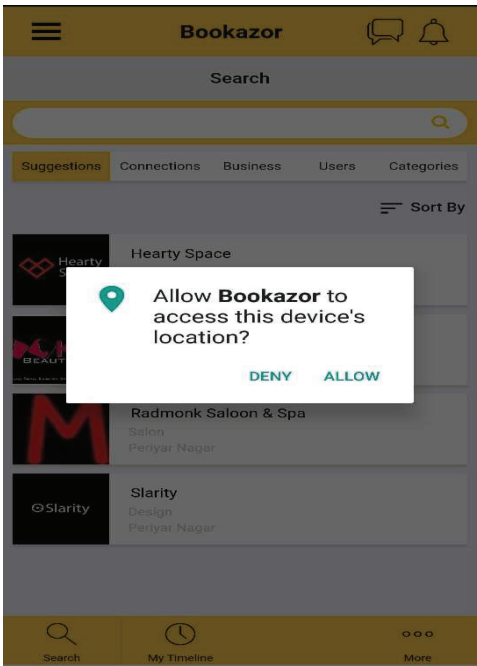


Fig1: Enabling the location

The login is available on the left pane, on choosing the login; the ‘create a new user’ screen is prompted to the user. The user enters his/her credentials like first name, last name and phone number. Register option is chosen if the user is signing up; if he/she is an existing user, then the sign in option is chosen for

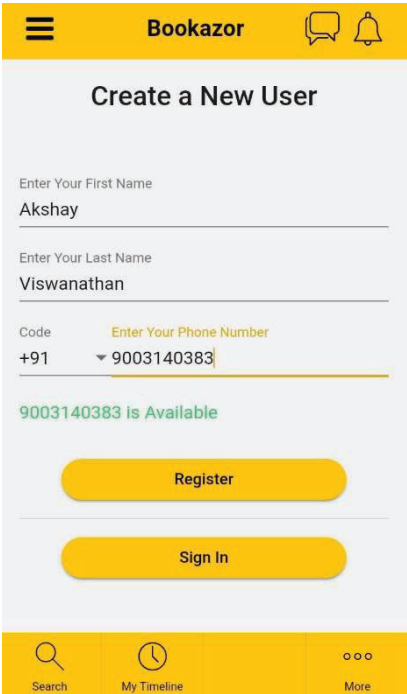


Figure 2: Login screen

When the register option is chosen, an OTP is generated and sent to the registered number for verification. The generated OTP is entered and as soon as the verify OTP is chosen the user is asked to enter the user name, mail id and password in the next screen. If the OTP is not received by the user, the user can choose “Resend OTP”.

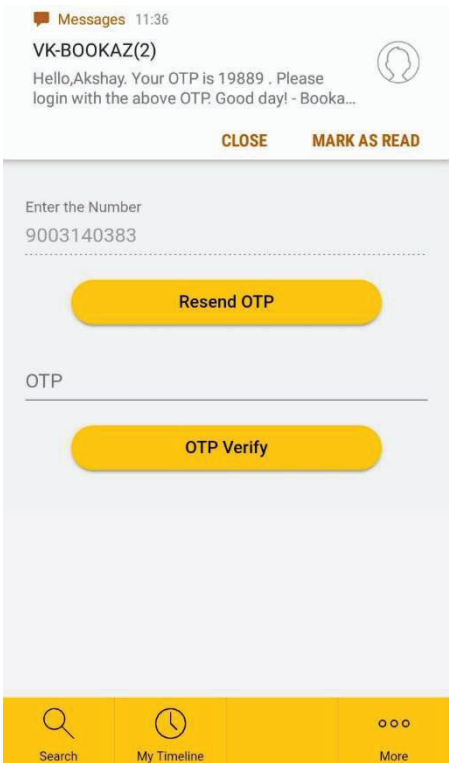


Figure 3 : OTP generation

After the OTP process, on clicking register, an automated mail is sent to the registered email id for verification. The user can proceed to use the application in their mobile after verifying their credentials by clicking link that is sent to their respective mail id.

“Your Email has been verified” denotes the acceptance of the user to use the application. The done option is chosen after verifying the mail id to use the application for further process.

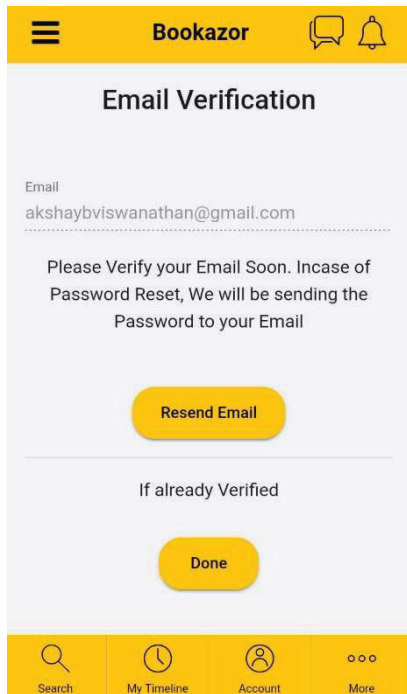


Figure 4: Email ID verification

3. 2. REGISTERING AND APPOINTMENT MODULE:

Once the user has logged in with their credentials, the user can register themselves with respect to their categories or if the registered user is not a part of any category he/she can view the category list and choose the category for which he/she needs an appointment to be scheduled. Figure 5 and 6 depicts the categories.

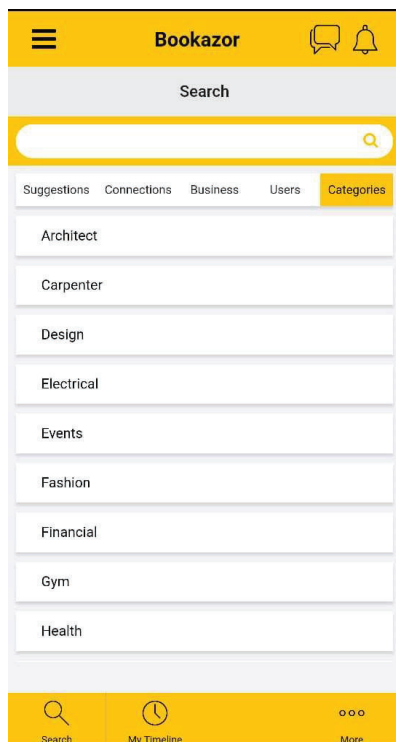


Figure 5: Category List

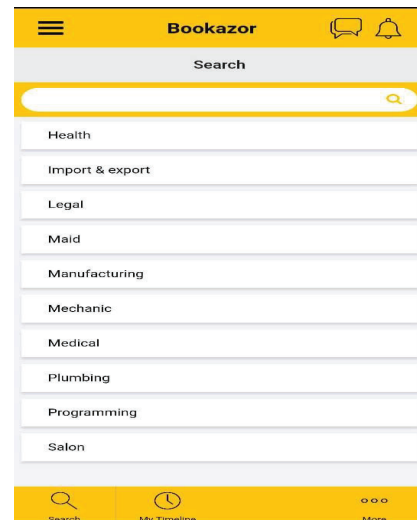


Figure 6: Category list

If the user wants to register their company, they must select the category to which their company belongs to and click the option “Create Business Profile” and enter the details like : Company Name, Photo of their brand, Category Name, Email, Country, City, Address, Pincode so that the customers can view their profile under both business tab and users tab. After viewing the profile the user may decide whether or not to schedule an appointment. The customer might want to schedule an appointment of his/her own desire and for that the category and the in the category suitable company/brand must be chosen and then there is a slider which will let the user to fix a time. The options available and unavailable changes with respect to the user’s/customer’s time request for the appointment. For example: Assume that there are few companies already registered in the category named Salon; ie, they have a business profile. The customers wanting to schedule an appointment must choose the category tab and select the salon. In the salon category, the user may choose any of the salons and schedule an appointment with respect to the availability. Figure 7 is the depiction of the illustrated example.

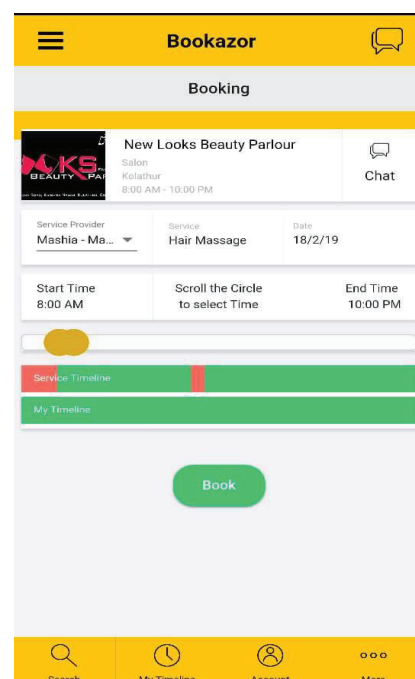


Figure 7: Booking

The circle on the slider bar helps the user to schedule an appointment with ease. By just sliding the bar, the user will be able

to easily schedule the timing for the appointment.. As soon as the user books an appointment, the request is processed and sent to the respective company and the user has to wait for the confirmation; until then the status remains “Pending”. The user has the privilege to cancel the appointment anytime. “My Timeline” option in the bottom pane helps the user to view their appointment details and its status. There is an extra feature which enables the user to chat with the company using the “chat” at the right corner. The chat option improves the interaction and communication between the customer and the company. The service provider lists the name of the stylists based on the pattern or style they are expertised at. The user can select their desired service type. The services option provides various services available with the cost. Each service has its own cost. The services can be bundled as a package and each package will be tagged with a price. The user can select any one of the services. To schedule for a particular service the date has to be chosen. The service timeline denotes the availability i.e; the service time of the company. The green colour denotes the time availability and the red colour denotes that the company is not available currently for serving the customer. My timeline is to denote the customer’s availability. Thus, the customer has to book an appointment only in the service time in which the company is available i.e; only with respect to the service time of the company. This example is for scheduling an appointment in a salon. The same process and procedure holds for other categories. The search helps in searching the needed category. The option named “more” new features can be suggested by the user and also if any new features/threads that includes incremental features will be hosted there.

3. BUSINESS, CONNECTIONS AND SUGGESTIONS:

The business, connection and suggestions tab are present in the welcoming screen itself, which enhances the usage of this application.

a) Business tab :

IV.

This tab enables the different companies to communicate amongst themselves which results in collaboration of ideas. The segregated individual companies come under a single category and this tab includes all the uncategorized business domains with regard to all the registered business profile and helps users and the companies themselves for promotion for easy appointment scheduling and booking.

b) Connection tab :

Using this tab the users can connect to any company and also the people who can have registered already. The connection request has to be given to the company or user that the one needs to connect. Once the request is accepted, the user can connect and share the ideas or communicate anything with regard to business or appointment.

c) Suggestions tab:

The suggestion tab helps the user to choose and opt for the right category by paging the users in the right direction. The suggestion tab is a general tab which shows the company belonging to all the categories based on various criteria. The criteria include: geographical location of the user, the user’s previous visits, search history.

4. CHATBOT :

Chatbot is the new feature that helps the user to stay in the right page of the application and any queries can be posted by the user which will be immediately answered by the appropriate respective team. This feature eliminates the chaos and confusion by directing the users in the right direction. This feature takes the queries of the user as input, understands user’s needs, provides valuable information, and increases the credibility.

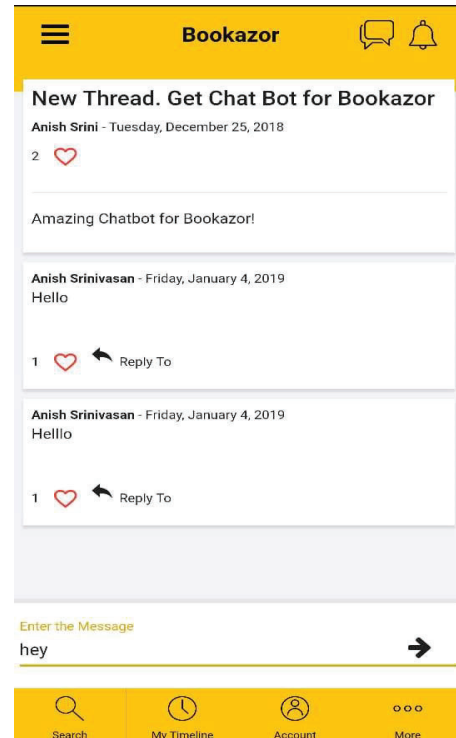


Figure 8 : Chatbot functionality

IV. DESIGN AND ARCHITECTURE

Angular is used as the front end of the app. When the user requests for a new appointment, the front end will send a request to the server. The cloud functions are enabled by the NodeJS that acts as the Backend Server. The Cloud Functions in the backend will check if there's any existing appointments overlapping with the requested appointment, if not overlapped, then the backend will create a new appointment. After the appointment is fixed, it sends a push notification to the Service Provider using FCM (Firebase Cloud Messaging). Once the service provider receives the notification, He/She will accept/deny the appointment using the front-end and the result will be updated back to the backend. The backend will push a notification to the user using FCM whether the Service Provider has accepted the booking or not. An SMS function will be triggered by the Cloud Function in case of updates like Booking Confirmation, Cancellation and it is handled by MSG91 API. The application does not make use of any huge algorithms rather it uses only modules for front end, back end, processing and scheduling. MSG91 API is used for authenticating users with OTP sent over SMS. Google Charts to visualize busy hours in real time. Ionic module to enable mobile ready applications.

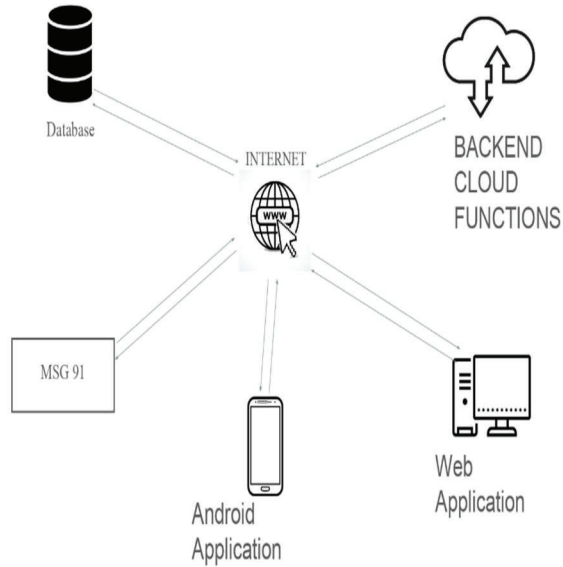


Figure 9: Architecture of the application

Web application/Mobile application acts as the front end where, the user has to create a profile for the organization comprising of location, details and branding or the profile is created by the user for booking an appointment in the already registered line of business. As and when the user profile is created it is stored in the database. The details are retrieved whenever required. The profile duplication is checked time and then. The next step of the procedure is scheduling. The duplicate search is done for the schedule request also. Thus the architecture is very simple where the input, process and output involves only the store and retrieve strategy. The store and retrieve process is done using the cloud functions. The cloud functions are operated via internet that includes both the store and retrieve process. Intimation to the users helps the users to acknowledge their appointments. To intimate the users regarding their appointment. Mytimeline option in the application is used by the users to view the appointment and for the reminder of the appointment and also to intimate that the user has fixed or scheduled for a particular servicetype with a corresponding service in a predefined category a message is sent to the user. It is like a push notification function that is sent by the Cloud function. This function is handled by the MSG91 API. The cloud functions are enabled by the Backend Server NodeJS which helps in easy retrieval and storage process. This server eliminates the tedious process of accounting and validating each profile of the registered user. Authentication and authorization security measures are taken prior to avoid any issues. Since it is ionic in nature, the user can use any front means to schedule an appointment: either via web application or via mobile application. The architecture aims at providing a framework that can be further enhanced in the future years if any incremental changes are required. All the afore mentioned modules are interconnected and they are provisioned by means of only internet medium. The needs to enable the internet option for access. Enabling the location provides further readiness for providing the user with the category of organizations that are located near the geographical location of the user.

V. CONCLUSION

We have given a sound and complete mobile and web application systematic procedure for enhancing the appointment booking process. To strive in this world of blooming technologies where each company has their own application for interacting with the customers; evolution and interfacing is required. The main disadvantage is that, the users have to download all the applications to schedule an appointment in the respective application. So to provide an easy access to users, Bookazor is introduced. It helps the user to download just one application where, the users can schedule an appointment in any category comprising of any company that has been previously registered in the application. It helps the business users also to communicate, interact and share ideas in the common forum which leads to greater scalability and flexibility. Many techniques, algorithms and applications have been separately developed for appointment booking that includes many tools as well. This application is cross functional in nature and so it can be used as both mobile application and a web application. The system is supported at a platform that is feasible with the existing technology. In this application, the process of booking an appointment is made simple and quick, all the services with respect to numerous divisions and units of various organizations is available under a single cover, it is timely in nature which records the history of user's appointments and helps the user with suggestions while reusing the application for scheduling. The need for this application will grow in this growing technical world as this application is a unified consistent platform for scheduling appointments comprising of various domains.

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